ANXIOLYTICS–2026

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ANXIETY

- UPLEASANT STATE OF TENSION, APPREHENSION OR UNEASINESS
- S/S
  - TACHYCARDIA
  - SWEATING
  - TREMBLING
  - PALPITATIONS
- TREATMENT IS REQUIRED FOR SEVERE, CHRONIC, DEBILITATING ANXIETY

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DEFINITIONS:

- **Anxiolytic/Sedative**:

Agent that relieves anxiety and exert a calming effect

- **Hypnotics**:

It produce sleep (encourage the onset and maintenance of state of sleep)

- Hypnotics have more CNS depressant actions than Sedatives.

SEDATIVE AUTONOMIC DRUGS:

- Antipsychotics
- Antihistamines
- Antidepressants (SSRIs)

They are sedatives but have marked effects on autonomic system

- Not produce General Anesthesia
- No Abuse liability.

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CLASSIFICATION:

Two Major Groups

- Barbiturates(Older)

- Benzodiazepines(Newer)

Others: Zolpidem, Buspirone

BENZODIAZEPINES

- Diazepam
- Chlordiazepoxide
- Flurazepam
- Lorazepam
- Triazolam
- Alprazolam

BARBITURATES

Phenobarbital

Thiopental

Secobarbitone

Pentobarbitone

Amobarbital

Butalbital

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CLSSIFICATION OF BENZODIAZIPINES

1. SHORT ACTING (3-8 hours)
TRIAZOLAM (5 hrs), MIDAZOLAM (4 hrs), OXAZEPAM, ALPRAZOLAM

2. INTERMEDIATE ACTING (10 -20 hours)
TEMAZEPAM (20hrs), ESTAZOLAM (19hrs),LORAZEPAM (20hrs)

3. LONG ACTING (1-3 days)
CLONAZEPAM(39hrs), DIAZEPAM (50hrs), CHLORDIAZEPOXIDE (30) ,FLURAZEPAM (47-100hrs), QUAZEPAM

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BARBITURATES

CLASSIFICATION

Long-acting (1-2 days)

- Phenobarbital
- Methylbarbitone

CLASSIFICATION

Short-acting (3-8 hrs)

- Secobarbitone
- Pentobarbitone
- Amobarbital
- Butalbital

Ultra-short-acting (20min)

- Thiopental

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QUESTION

A patient with generalized anxiety disorder requires a benzodiazepine with a very long duration of action because of less frequent dosing.

A. Triazolam
B. Oxazepam
C. Midazolam
D. Diazepam

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PHARMACOKINETICS:

Benzodiazepines:

- Oral absorption
- Lipophilic
- Penetrate CNS
- Cross placenta : Depress CNS of newborn
- Hepatic Metabolism (Microsomal Enzymes P-450/ Glucuronidation)
- Excretion through Kidney
- Metabolized by hepatic microsomal enzymes and Form Active Metabolites

Like:

Chlordiazepoxide, diazepam

QUESTION

- An elderly patient with liver disease needs a benzodiazepine for anxiety. The physician avoided a drug that undergoes metabolism by hepatic microsomal enzyme and have active metabolites.

A. Chlordiazepoxide
B. Oxazepam
C. Temazepam

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MECHANISM OF ACTION:

- They Bind To Gaba A Receptor
- It Is An Ionotropic Transmembrane Protein That Functions As Chloride Channel, Is Activated By The Inhibitory Neurotransmitter Gaba.
- Three Subunits Are Important: Alpha, Beta And Gamma
- Benzodiazepines Bind At The Interface of The Alfa And Gamma Subunit of Gaba

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- Resulting In Influx of Chloride

MECHANISM OF ACTION:

- Benzodiazepines increases frequency of channel opening
- Benzodiazepines Potentiate Gaba-ergic Inhibition Via Membrane Hyperpolarization
- They Enhance Gabaergic Effects Without Directly Activating Gaba Receptor

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QUESTION

A 25-year-old anxious man is prescribed diazepam before a dental procedure. The drug acts by enhancing inhibitory neurotransmission in the CNS.

A. Blocks NMDA receptors
B. Increases the frequency of chloride channel opening mediated by GABA
C. Increases the duration of chloride channel opening mediated by GABA
D. Inhibits serotonin reuptake

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BENZODIAZEPINE RECEPTOR LIGANDS:

- Agonists
- Antagonists:Flumazenil
- Inverse Agonist:Negative allosteric modulator of GABA receptor e,g: Beta Carboline

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ORGAN SYSTEM EFFECTS:

- SEDATION
- ANXIOLYTIC
- HYPNOSIS
- GENERAL ANESTHESIA
- ANTICONVULSANT
- MUSCLE RELAXATION
- RESPIRATORY AND CARDIOVASCULAR DEPRESSION

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ACTIONS

1. REDUCTION OF ANXIETY (LESS TOLERANCE)

- LOW DOSES (2-10 mg): ANXIOLYTIC
- NEURONS WITH ALFA-2 SUBUNIT IN GABA-A RECEPTOR MEDIATED

2. SEDATIVE HYPNOTIC:

- ALFA-1 GABA RECEPTOR: HYPNOTIC EFFECT

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3. ANTEROGRADE AMNESIA:

- ALFA-1 GABA RECEPTOR MEDIATED

ACTIONS

4.ANTICONVULSANT:ALFA-1 GABA A MEDIATED

5. MUSCLE RELAXANTS

- RELAX SPASTICITY OF SKELETAL MUSCLE
- ALFA-2 GABA A RECEPTOR MEDIATED: PRESYNAPTIC INHIBITION IN THE SPINAL CORD

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ANTIANXIETY

- Anxiety Disorders
- Panic Disorders
- Gad
- Social Anxiety Disorders
- Performance Anxiety Disorders
- Extreme Phobias
- Anxiety Related To Depression And Schizophrenia

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- It Is Addictive Limited Use
- Antianxiety Effect Is Less Subject To Tolerance Than Sedative And Hypnotic Effects

QUESTION

A 40-year-old woman has been taking diazepam nightly for several months. She develops tolerance to its sedative effect but continues to experience relief of anxiety.

A. Sedation
B. Hypnosis
C. Anticonvulsant action
D. Anxiolytic effect

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AMNESIA

- SHORT ACTING AGENTS ARE USED AS PREMEDICATION (PROVIDING SEDATION DECREASES ANXIETY BEFORE PROCEDURES) FOR ENDOSCOPY, DENTAL PROCEDURES AND ANGIOPLASTY
- PRECONCIOUS SEDATION
- PATIENT IS RECEPTIVE TO THE INSTRUCTIONS
- MIDAZOLM IS USED

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QUESTION

A nervous patient is scheduled for an upper gastrointestinal endoscopy. He is given a benzodiazepine to relieve anxiety and produce amnesia before the procedure.

A. Midazolam
B. Phenobarbital
C. Buspirone
D. Zolpidem

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SEIZURES

- DRUG OF CHOICE FOR STATUS EPILEPTICUS: DIAZEPAM AND LORAZEPAM
- AS AN ADJUNCT FOR TREATMENT OF EPILEPSY: CLONAZEPAM
- ALCOHOL WITHDRWAL SYNDROME RELATED SEIZURES: CHLORDIAZEPOXIDE, CLORAZEPATE, DIAZEPAM, LORAZEPAM AND OXAZEPAM

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MUSCULAR DISORDERS

- SKELETAL MUSCLE SPASMS ---CEREBRAL PALSY AND MUTIPLE SCLEROSIS INDUCED SPASTICITY: **DIAZEPAM**

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QUESTION

A 34-year-old woman with multiple sclerosis develops painful muscle spasms and increased muscle tone. Her physician prescribes diazepam.

A. Antidepressant action
B. Anticonvulsant action
C. Skeletal muscle relaxation
D. Sedation only

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EFFECT ON SLEEP:

- Sedative And Calming Effects
- Decrease Latency to sleep onset
- Decrease REM Sleep
- Decrease Slow Wave Sleep
- Increase Stage 2 Of NREM (light sleep)
- To Avoid Residual Sedation (Hangover) Triazolam Preferred (People Having Difficulty In Falling Asleep) But Increases Risk Of Withdrawal And Rebound insomnia
- Patient Experiences Frequent Awakenings Temazepam Is Preferred

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INDICATIONS:

- ANXIETY STATES: (CLONAZEPAM, LORAZEPAM, DIAZEPAM)
- PANIC DISORDER: (ALPRAZOLAM)
- SLEEP DISORDERS: (TEMAZEPAM, TRIAZOLAM)
- AMNESIA (MIDAZOLAM)
- MUSCULAR DISODERS (DIAZEPAM)
- PRE-ANESTHETIC PROCEDUREs (ENDOSCOPY,BRONCHOSCOPY)---MIDAZOLAM
- MANIA (INITIAL MANAGEMENT)
- STATUS EPILEPTICUSSEIZURE--DIAZEPAM,LORAZEPAM)
- ALCOHOL WITHDRAWAL SYNDROME (CHLODIAZEPOXIDE, CLORAZEPATE, DIAZEPAM, LORAZEPAM,OXAZEPAM)
- MAJOR DEPRESSIVE EPISODE (ALPRAZOLAM)

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QUESTION

- A 50-year-old man with chronic alcohol use develops tremors, agitation, and sweating after abruptly stopping alcohol intake.

- A. Phenobarbital
  B. Diazepam
  C. Buspirone
  D. Ramelteon

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CONTRAINDICATIONS

- LIVER DISEASE
- ALCOHOL
- ACUTE ANGLE CLOSURE GLUCOMA

ADVERSE EFFECTS:

- ATAXIA
- DROWSINESS
- CONFUSION
- COGNITIVE IMPAIRMENT
- RESIDUAL SEDATION (HANGOVER)
- ANTEROGRADE AMNESIA(INABILITY TO REMEMBER THE EVENTS DURING DRUG'S ACTION)
- TOLERANCE
- DEPENDENCE
- WITHDRAWAL SYNDROME (Confusion, anxiety, agitation, insomnia, seizures severe with triazolam)
- RESPIRATORY DEPRESSION (LETHAL DOSE)

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QUESTION

A 45-year-old man takes a long-acting benzodiazepine at bedtime for insomnia and reports drowsiness and impaired concentration the next morning.

A. Rebound insomnia
B. Dependence
C. Hangover effect
D. Paradoxical excitation

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FLUMAZENIL:

- GABA RECEPTOR ANTAGONIST
- RAPIDLY REVERSES THE EFFECTS OF BENZODIAZIPINES
- IV, HALF LIFE 1 HOUR
- INDICATION:
  - BENZODIAZEPIE OVERDOSE
  - ANESTESIA SPEEDY RECOVERY
- BLOCKS ONLY ACTIONS OF BENZODIAZEPINES
- CAN PRECIPITATED ABSTINENCE SYNDROME
- ADRs: DIZZINESS, NAUSEA VOMITING

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QUESTION

A young woman is brought to the emergency department after ingesting a large number of sleeping pills. She is drowsy but breathing adequately, and benzodiazepine overdose is suspected.

A. Naloxone
B. Physostigmine
C. Flumazenil
D. Atropine

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BUSPIRONE

- Affinity for Serotonin (5-HT1A) Receptors)---Partial agonist ---Primary action
- Affinity for D2 receptors and 5HT-2A (Antagonistic effect)
- Slow onset of action
- Useful for chronic treatment of gad
- ADRs: headache, dizziness, nervousness, nausea, lightheadedness
- Dependance unlikely
- Minimal psychomotor, cognitive and sedative
- Don't potentiate CNS depression of alcohol
- Not used in acute anxiety
- Lacks anticonvulsant effect
- Lacks muscle relaxant effect

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APPROVED ANTIDEPRESSANTS FOR GAD

- SSRIs

Escitalopram, Paroxetine

- SNRIs

Venlafaxine or Duloxetine

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BARBITURATES

- WIDELY REPLACED NOW BY BENZODIAZEPINES BECAUSE OF INDUCING TOLERANCE, DRUG-METABOLIZING ENZYMES, PHYSICAL DEPENDENCE & SEVERE WITHDRAWAL SYMPTOMS & COMA AT TOXIC DOSES

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- SOME ARE STILL USED TO INDUCE ANESTHESIA

BABITURATES CLASSIFICATION

Long-acting (1-2 days)

- Phenobarbital
- Methylbarbitone

Short-acting (3-8 hrs)

- Secobarbitone
- Pentobarbitone
- Amobarbital
- Butalbital

Ultra-short-acting

(20min)

- Thiopental

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QUESTION

A patient requires rapid induction of anesthesia for emergency surgery. An ultra-short-acting barbiturate is selected intravenously.

A. Phenobarbital
B. Pentobarbital
C. Thiopental
D. Secobarbital

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BARBITURATES

- ABSORBED ORALLY AND DISTRIBUTED WIDELY THROUGHOUT BODY, RAPIDLY CROSS PLACENTA
- MOSTLY METABOLIZED VIA HEPATIC ENZYMES
- APPEAR IN URINE AS GLUCORONIDE CONJUGATES
- SOME ARE EXCRETED UNCHANGED

Mechanism of action

- INTERACT WITH $GABA_A$ RECEPTORS ENHANCING GABANERGIC TRANSMISSION
- BINDING SITE DIFFERENT FROM BENZODIAZEPINES
- POTENTIATE GABA ACTION AND CHLORIDE ENTRY INTO NEURON BY PROLONGING THE DURATION OF CHLORIDE CHANNEL OPENING
- CAN ALSO BLOCK EXCITATORY GLUTAMATE RECEPTORS

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MECHANISM OF ACTION

- **PROLONGING THE DURATION OF CHLORIDE CHANNEL OPENING**
- **BLOCK EXCITATORY GLUTAMATE RECEPTORS**

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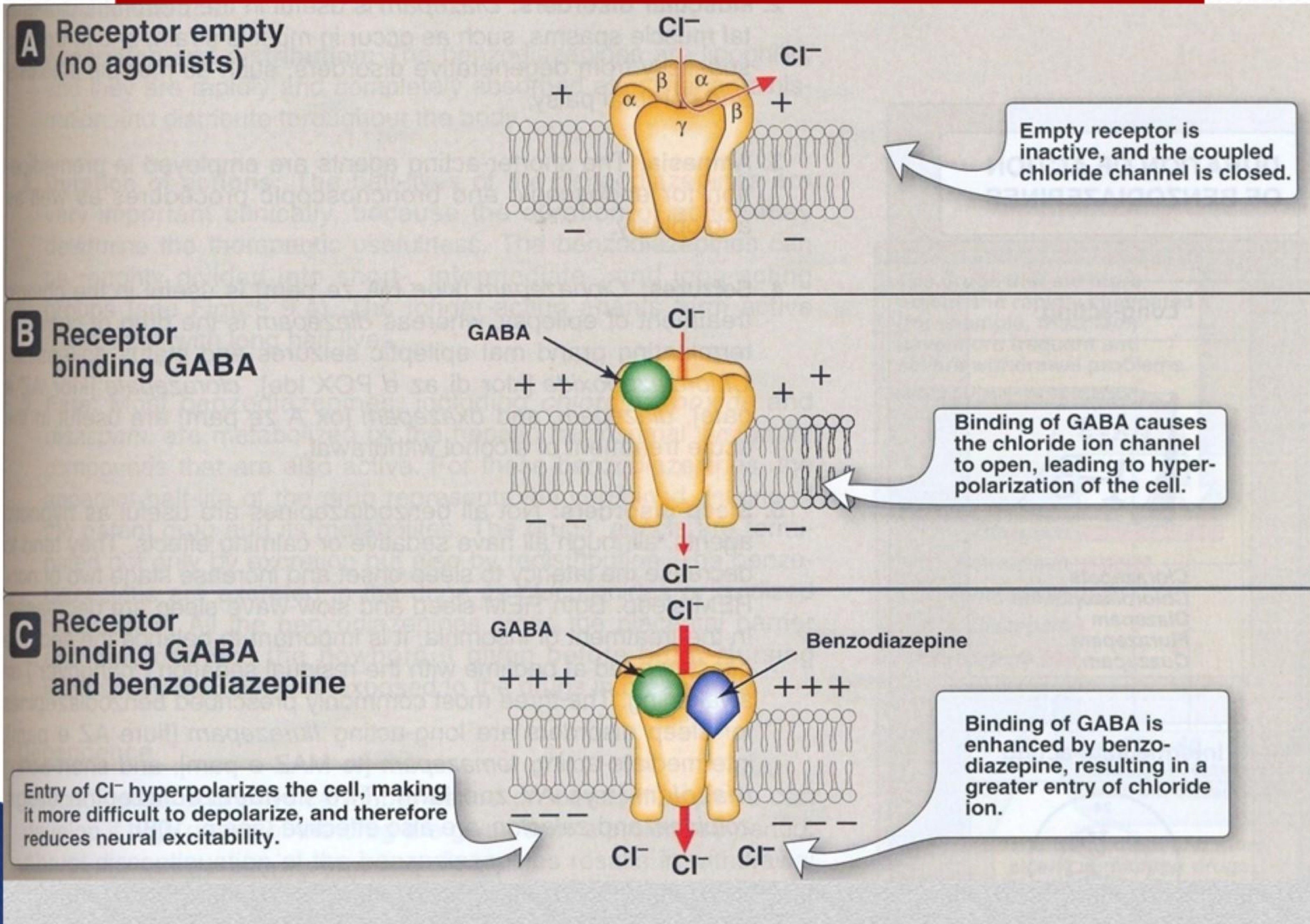

A Receptor empty (no agonists)
Cl⁻
Cl⁻
β α
α β
γ
+
+
−
−
Empty receptor is inactive, and the coupled chloride channel is closed.
B Receptor binding GABA
GABA
Cl⁻
+ +
+ +
− −
− −
Cl⁻
Binding of GABA causes the chloride ion channel to open, leading to hyper-polarization of the cell.
C Receptor binding GABA and benzodiazepine
GABA
Cl⁻
Benzodiazepine
+++
+++
− − −
− − −
Cl⁻ Cl⁻ Cl⁻
Binding of GABA is enhanced by benzo-diazepine, resulting in a greater entry of chloride ion.
Entry of Cl⁻ hyperpolarizes the cell, making it more difficult to depolarize, and therefore reduces neural excitability.

QUESTION

A patient with severe migraine receives a combination product containing an analgesic and butalbital. The physician explains that barbiturates differ from benzodiazepines in their effect on GABA receptors.

A. They increase the frequency of chloride channel opening.
B. They increase the duration of chloride channel opening.
C. They block dopamine receptors.
D. They inhibit serotonin reuptake.

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ACTIONS

- SEDATION
- HYPNOSIS
- SUPRESS REM SLEEP
- P450 INDUCER
- TOLERANCE
- PHYSICAL DEPENDANCE
- SEVERE WITHDRAWAL SYNDROM
- REPIRATORY DERESSION
- NO ANALGESIC EFFECT
- LOW TI
- Hangover
- CONTRAINDICATED IN ACTUE INTERMITTENT PORPHYRIA

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Actions

- AT LOW DOSES: SEDATION (CALMING EFFECT, REDUCING EXCITEMENT)
- AT HIGH DOSES: HYPNOSIS FOLLOWED BY ANESTHESIA AND FINALLY COMA AND DEATH
- NO ANALGESIC EFFECT
- TOLERANCE MAY DEVELOP ON CHRONIC USE

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ACTIONS

- SUPPRESS THE HYPNOTIC CHEMORECEPTOR RESPONSE TO CO_2
- AT HIGH DOSES: RESPIRATORY DEPRESSION AND DEATH
- INDUCES P450 HEPATIC MICROSOMAL ENZYMES
- SUPRESS REM SLEEP

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THERAPEUTIC USES

ANTICONVULSANT:

- PHENOBARBITAL LONG-TERM MANAGEMENT OF TONIC-CLONIC SEIZURES, STATUS EPILEPTICUS

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- DRUG OF CHOICE IN CHILDREN FOR FEBRILE SEIZURES

ANTICONVULSANT

- FEBRILE SEIZUR(IN CHILDREN AS AN ALTERNATIVE AS IT DEPRESSES COGNITIVE DEVELOPMENT)
- STATUS EPILEPTICUS AS AN ALTERNATIVE

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THERAPEUTIC USES

ANESTHESIA:

- **ULTRA-SHORT ACTING EG: THIOPENTAL USED I/V TO INDUCE ANESTHESIA**

ANXIETY:

- **USED AS MILD SEDATIVES TO RELIEVE ANXIETY, NERVOUS TENSION, INSOMNIA**
- **MIGRAINE HEADACHE**

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(CAFFIEN +ACTAAMIPNOPHINE OR CAFFEIN + ASPIRIN)

DRUG INTERACTIONS

- INDUCE P450 SYSTEM, MAY DECREASE DURATION OF ACTION OF DRUGS METABOLIZED BY THIS SYSTEM
- INCREASES PORPHYRIN SYNTHESIS; CONTRAINDICATED IN ACUTE INTERMITTENT PORPHYRIA
- SUDDEN WITHDRAWAL MAY RESULT IN TREMORS, ANXIETY, WEAKNESS, RESTLESSNESS, NAUSEA, AND VOMITING, SEIZURES, DELIRIUM, AND CARDIAC ARREST

ADVERSE EFFECTS

- NAUSEA
- DROWSINESS
- IMPAIRED CONCENTRATION
- MENTAL AND PHYSICAL SLUGGISHNESS
- DRUG HANGOVER
- TOLERANCE
- PHYSICAL DEPENDANCE
- SEVERE WITDRAWAL SYNDROME
- DDI

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POISONING

- MAY RESULT IN DEATH AT HIGH DOSES
- CAUSES SEVERE RESPIRATORY & CARDIOVASCULAR DEPRESSION RESULTING IN SHOCK-LIKE CONDITION WITH SHALLOW, INFREQUENT BREATHING
- TREATMENT INCLUDES ARTIFICIAL RESPIRATION AND GASTRIC LAVAGE IF DRUG IS TAKEN RECENTLY; HEMODIALYSIS MAY BE REQUIRED IF TAKEN IN LARGE QUANTITIES
- IN PHENOBARBITAL POISONING, ALKALINIZATION OF URINE HELPS IN ELIMINATION

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QUESTION

A 28-year-old man takes several tablets of phenobarbital for insomnia and is brought to the emergency department with marked CNS depression and reduced responsiveness.

A. CNS depression progressing to coma
B. Extrapyramidal symptoms
C. Hypertensive crisis
D. Serotonin syndrome

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NONBNZODIAZIPINE DRUGS

- ZOLPIDEM
- ZALEPLON
- ESZOPICLONE

<u>ZOLPIDEM</u>

- BINDS TO BZ 1 RECEPTORS
- NO ANTICONVULSANT AND MUSCLE RELAXANT EFFECTS
- MINIMAL REBOUND INSOMNIA
- ANTEROGRADE AMNESIA
- DO NOT ALTER SLEEP STAGES
- USED IN IINSOMNIA

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MELATONIN RECEPTOR AGONIST

- **RAMELTEON**

- M1 AND M2 melatonin receptor agonist
- Induce and promote sleep
- Minimal potential for abuse
- No dependence
- No withdrawal symptoms
- ADRs ---Dizziness, Fatigue, Somnolence

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THANK YOU